

TrES-3b

R-1753

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210504 · created 2026-08-02T20:06:33+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459338.8187 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.0095
Transit depth (Rp/Rs)^2	2.02 +/- 0.27 [%]
Orbital Inclination (inc)	86.85 +/- 0.73
Airmass coefficient 1 (a1)	1.278 +/- 0.013
Airmass coefficient 2 (a2)	-0.217 +/- 0.0065
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	None
Optimal Aperture	3.66
Optimal Annulus	11.98
Transit Duration (day)	0.0792 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.0095 vs 0.16309 (deviation 2.22σ)
Duration fit vs published	0.0792 d vs 0.0567 d (deviation 11.27σ)
Mid-time O–C	3.0 min
Depth SNR	14.9
Residual scatter	1.06 %

Light curve

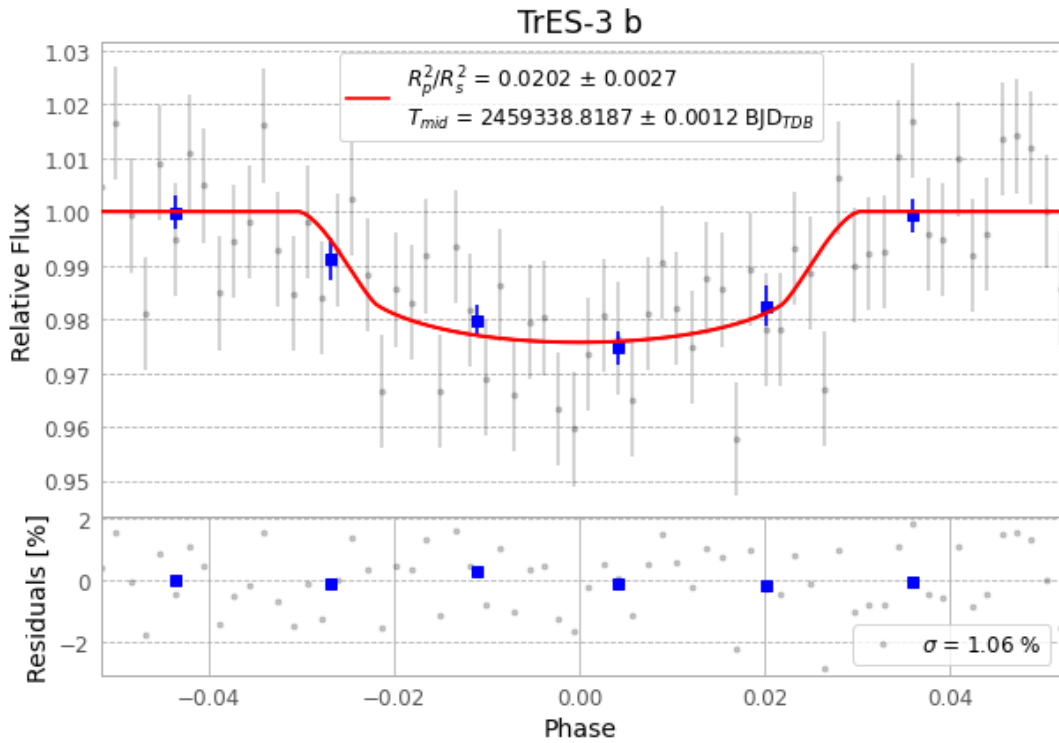


Figure 1 — FinalLightCurve_TrES-3 b_2021-05-03.png (SHA-256 906de89a06180dbc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ae1fd49ba98f902...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-3210504055712.FITS → TRES-3210504091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210504.log (28c5869f87e5...), FinalLightCurve_TrES-3 b_2021-05-03.png (906de89a0618...), FinalLightCurve_TrES-3 b_2021-05-03.csv (5495b36b9472...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 20:06Z.